Question			Answer	Marks	AOs	Guidance	
7	(a)		**DR** Midpoint of AB is (3, 1) Centre C of the circle is (3, 1) and radius $\sqrt{(7-3)^2+(-2-1)^2}=5$ So circle is $(x-3)^2+(y-1)^2=25$	**B1** **M1** **M1** **A1** **[4]**	**3.1a** **3.1a** **1.1b** **1.1b**	soi Attempt to find length of AB, AC or BC Uses their midpoint and radius (do not allow for diameter used) Need not be simplified	
7	(b)		**DR** Crosses $y=2x+5$ where $(x-3)^2+(2x+5-1)^2=25$ $5x^2+10x=0$ giving $x=-2, 0$ So points are (–2, 1) and (0, 5)	**M1** **A1** **A1** **[3]**	**1.1b** **1.1b** **1.1b**	Substituting $y=2x+5$ and attempting to collect terms oe Both values correct Correct *y* coordinates FT their *x*-coordinates	Allow for a quadratic solved BC providing it is seen in form $ax^2+bx=0$ or $ay^2+by+c=0$
7	(c)		**DR** AQ = $\sqrt{2}$ and BQ = $\sqrt{7^2+7^2}=7\sqrt{2}$ Triangle ABQ has a right angle at Q (angle in a semicircle) So area of triangle is $\frac{1}{2}\times AQ\times BQ$ Area = 7	**M1** **M1** **A1** **[3]**	**3.1a** **2.1** **1.1b**	Attempt to find two lengths to be used in their area calculation (excluding AB) Correct method for finding the area FT their Q	Note QAB=81.9° and QBA=8.1°